

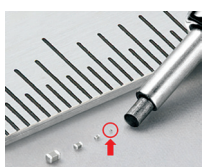
NEW PRODUCTS

TECHNOLOGY

COMPONENTS

Multilayer ceramic capacitors

Murata Manufacturing Co. Ltd has introduced the world's smallest multilayer ceramic capacitors (MLCCs),



measuring only 0.16mm × 0.08mm, achieving a 75% volume reduction over its previous model. These

MLCCs support greater functionality while minimising space. Devices like smartphones can incorporate up to 1000 MLCC units.

Murata Manufacturing Co. Ltd
<https://www.murata.com>

High-bandwidth current sensors

Allegro's new CT455 and CT456 current sensors offer precise, low-noise measurements with high bandwidth,



enhancing energy efficiency and reducing space and cost in high-power designs.

Built on XtremeSense technology, these compact sensors are ideal for engineers in AI data centres, automotive powertrains, and power conversion systems, supporting 80A to 800A current ranges with high precision. The CT455 provides linear analogue output with two field ranges, while the CT456, with a 6mT field range, suits high-accuracy applications.

Allegro MicroSystems
<https://www.allegromicro.com/en>

GaN switcher IC

Power Integrations has launched its 1700V InnoMux-2 IC, featuring

cutting-edge GaN technology and PowiGaN innovation. This IC advances GaN device voltage capabilities from 1250V to 1700V,



allowing efficient operation from a 1000V DC bus and achieving over 90% efficiency. Designed

for single-stage, multi-output power supplies, it supports up to 60W across three outputs. Key features include 10% efficiency gain, fewer components, and under 50mW no-load consumption. Ideal for automotive chargers, solar inverters, three-phase meters, and industrial systems.

Power Integrations
<https://www.power.com>

MOSFET and IGBT gate drivers

Littelfuse's new IX4341 and IX4342 dual 5-amp low-side MOSFET gate drivers expand the IX434x series,



delivering efficient MOSFET control for diverse applications. With peak

drive currents of 1.5A to 30A, and compatibility with TTL and CMOS logic, these drivers suit switch-mode power supplies, DC-DC converters, motor controllers, and more. Fast 16ns propagation delays, 7ns rise/fall times, and low output impedance make them ideal for high-frequency tasks.

Littelfuse, Inc.
<https://www.littelfuse.com>

Energy-saving microcontrollers

Renesas Electronics Corporation, Japan, has introduced the RX261 and RX260 microcontroller (MCU) series.



Powered by a 64MHz RXv3 core, these MCUs operate at

just 69µA/MHz in active mode and 1µA in standby, providing 2.5 times the processing capability with 25% lower active current. Enhanced touch IP ensures noise immunity and water resistance, making them perfect for appliances and outdoor devices. Integrated security features like Secure IP and flash protection enhance secure boot and firmware updates.

Renesas Electronics Corporation
<https://www.renesas.com/en>

Isolated gate drivers

STMicroelectronics has launched the STGAP3S series of advanced galvanically isolated gate drivers for high-performance motor drives and



power control in diverse industries, including air-conditioning, home appliances, factory

automation, and energy systems like EV charging, solar inverters, and energy storage. These drivers support SiC MOSFETs and IGBTs, offering reinforced capacitive isolation with 9.6kV transient isolation and 200V/ns CMTI, enhancing reliability and protection. The series provides 10A and 6A options, with UVLO, desaturation protection, and an integrated Miller clamp for optimised, safe operation.

STMicroelectronics
<https://www.st.com>